# Predictive Modeling & Customer Segment Analysis Project: Marketing for retail Wine

Capstone Business Analytics Project —
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QMB 4302
8 December 2025

# Problem Goal & Statement

Goal Statement: To evaluate how customer demographics and status variables influence wine purchasing patterns to determine the company's ideal target audience.

Significance of the Problem: Wine is a premium luxury product. By identifying the exact demographic profiles that drive high wine sales, the business can move away from wasteful, broad marketing and execute highly optimized, tailored promotions that maximize profit margins and marketing efficiency.

# Variable Mapping

Dependent Variable (Target): * MntWines (Amount spent on wine over the last 2 years)
Independent Variables (Predictors):

- Income (Yearly household income)
- Kidhome / Teenhome (Family size / number of dependents)
- Education (Graduation, Master, PhD, etc.)
- Age (Calculated from customer birth year)
- Marital_Status (Single vs. Married/Together)

# Hypothesis Questions

1) Does Income affect wine spending?
2) Does Family Size lower wine spending?
3) Does Education drive up wine spending?
4) Does Age matter?
5) Does Marital Status make a difference?

# Baseline Descriptive Statistics

| Variable | Count | Mean | Median | Std Dev | Min | Max |
|---|---|---|---|---|---|---|
| Total_Spend ($) | 2216 | 605.8 | 396 | 602.1 | 5 | 2525 |
| Total_Purchases | 2216 | 12.5 | 12 | 7.2 | 0 | 44 |
| Income ($) | 2216 | 52,247 | 51,381 | 21,269 | 1730 | 115,000 |
| Age (Years) | 2216 | 45.1 | 45 | 11.8 | 17 | 67 |
| Recency (Days) | 2216 | 49.1 | 49 | 28.9 | 0 | 99 |
| Dependents | 2216 | 0.95 | 1 | 0.75 | 0 | 3 |

# Baseline Descriptive Statistics- Pivot Table

| Variable | Sum of Mean | Average of Median |
|---|---|---|
| Age (Years) | 45.1 | 45 |
| Dependent Variables | | |
| Dependents | 0.95 | 1 |
| Income ($) | 52247 | 51381 |
| Independent Variables | | |
| Recency (Days) | 49.1 | 49 |
| Total_Purchases | 12.5 | 12 |
| Total_Spend ($) | 605.8 | 396 |
| **Grand Total** | **52960.45** | **8647.333333** |

# Household Size Vs Luxury Wine Spend SQL data insights

| Family Size | Avg.Income | Avg. Wine Spend | Total Customers |
|---|---|---|---|
| 0 (High Value) | $65,677 | $487.65 | 632 |
| 1 | $47,712 | $268.98 | 1,115 |
| 2 | $44,612 | $142.15 | 416 |
| 3 | $46,677 | $161.34 | 50 |

Link to SQL Data Insights: Marketing-Data-SQL-Data-Analysis/Marketing_Campaign .csv at main · shanayulett/Marketing-Data-SQL-Data-Analysis

# Technical Execution & Statistical Modeling

## Question 1: Income Level

- **Hypothesis:** * *$H_0$:* There is no relationship between income and the amount spent on wine.
    - *$H_1$:* Higher-income consumers are likely to spend more on wine.
- **Statistical Result: Reject $H_0$.** Income proved to be the **strongest predictor** in the model. High-income customers spend significantly more on wine, validating that purchasing power directly increases lifestyle luxury spending.

## Question 2: Family Size

- **Hypothesis:** * *$H_0$:* There is no relationship between family size and the amount spent on wine.
    - *$H_1$:* Wine purchases are significantly higher among smaller family sizes.
- **Statistical Result: Reject $H_0$.** The data confirmed a moderate negative effect. Larger families (more children/teenagers at home) spend less on wine due to increased financial responsibilities towards household necessities.

# Baseline Descriptive Statistics

### Question 1: Income Level

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### Question 2: Family Size

- **Hypothesis:** * *$H_0$:* There is no relationship between family size and the amount spent on wine.
    - *$H_1$:* Wine purchases are significantly higher among smaller family sizes and single households.
- **Statistical Result: Reject $H_0$.** The data confirmed a moderate negative effect. Larger families (more children/teenagers at home) spend less on wine due to increased financial responsibilities towards household necessities.

### Question 3: Education Level

- **Hypothesis:** * *$H_0$:* Education level does not significantly affect wine spending.
    - *$H_1$:* Higher education levels are associated with increased wine spending.
- **Statistical Result: Reject $H_0$.** Education had a meaningful, moderate positive impact. Higher academic achievements correspond to higher wine sales, often linked to higher disposable income and lifestyle choices.

# Baseline Descriptive Statistics Cont

## Question 4: Customer Age

- **Hypothesis:** * *$H_0$:* Age does not significantly affect wine spending.
  - *$H_1$:* As age increases, the amount spent on wine increases.
- **Statistical Result: Reject $H_0$ with caveat.** Age was found to be statistically significant, but it accounted for very little real-world variance in wine spending compared to income. It is a minor influencing factor.

## Question 5: Marital Status

- **Hypothesis:** * *$H_0$:* There is no significant difference in wine spending between married and single customers.
  - *$H_1$:* Single customers spend more on wine than married customers.
- **Statistical Result: Reject $H_0$.** The Independent Samples T-Test revealed a statistically significant difference ($p\text{-value} = 0.015$). Single customers have an average wine spend of **$250**, whereas married customers average **$180**, showing that single consumers display more individual spending freedom for luxury items.

# Predictive Modeling Evaluation

- To help predict future wine spending, two separate modeling methods were used

1) Multiple Linear Regression:

   -Provided straightforward equations and excellment interpretability regarding how each baseline factor shifts sales, but it struggled to adapt to complex, non-linear human shopping habits.

2) Decision Tree Regression:

   -The decision tree was a better option since it successfully splits customers into different demographic categories that captured the complex, interacting patterns of human behavior the linear model missed which helped achieved a higher prediction.

# Executive Report & Business Recommendations

By translating these model metrics and statistical milestones into corporate strategy, the business should deploy the following actions:

- **Prioritize High-Income Segments:** Direct core luxury wine marketing campaigns explicitly at higher household income brackets, as income is the primary engine driving wine consumption.
- **Target Single Households and Small Families:** Design tailored, single-focused promotions or optimized product package sizes for single customers and households with zero dependents, capitalizing on their higher financial freedom.
- **Filter via Academic Benchmarks:** Utilize customer education levels as a strategic profiling filter to refine target lists for premium wine club invitations or high-tier distributions.
- **Deploy the Decision Tree Model:** Integrate the Decision Tree model into the customer relationship management (CRM) platform to screen database leads, accurately forecasting which prospects will yield the highest wine spending before launching active campaigns.